

Matrix Theory: Lecture 13

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The Moore-Penrose Inverse

Def. The matrix X is called a **Moore-Penrose pseudoinverse** of the matrix A if

- ① $AXA = A$,
- ② $XAX = X$,
- ③ Both XA and AX are Hermitian matrices, thus

$$(XA)^H = XA, (AX)^H = AX.$$

Remark: If A has an inverse then

- $AA^{-1}A = A$,
- $A^{-1}AA^{-1} = A^{-1}$,
- Both $A^{-1}A = I$ and $AA^{-1} = I$ is Hermitian.

Thus A^{-1} is a Moore-Penrose pseudoinvers.



The Moore-Penrose Inverse

Theorem 15.4

If Σ is an invertible square matrix then the only Moore-Penrose pseudoinverse of the block matrix

$$S = \left(\begin{array}{c|c} \Sigma & O \\ \hline O & O \end{array} \right)$$

(some zero blocks may be absent) is the block matrix (of suitable size)

$$S^+ = \left(\begin{array}{c|c} \Sigma^{-1} & O \\ \hline O & O \end{array} \right).$$



The Moore-Penrose Inverse

Theorem 15.5

If A^+ is a Moore-Penrose pseudoinverse for the $m \times n$ matrix A and U is a unitary matrix of size m , then $A^+ U^H$ is a Moore-Penrose pseudoinverse for the matrix UA .

Similarly, if V is a unitary matrix of size n then VA^+ is a Moore-Penrose pseudoinverse for the matrix AV^H .

If one of them is unique the two others are also unique.



The Moore-Penrose Inverse

Theorem 15.6

Every $m \times n$ matrix A has a unique Moore-Penrose pseudoinverse X which we denote A^+ . If $A = USV^H$ is its SV-decomposition then

$$A^+ = VS^+U^H,$$

where

$$S^+ = \left(\begin{array}{cccc|c} \sigma_1^{-1} & 0 & \cdots & 0 & \\ 0 & \sigma_2^{-1} & \cdots & 0 & \\ \cdots & \cdots & \cdots & \cdots & \\ 0 & 0 & \cdots & \sigma_r^{-1} & \\ \hline & & & & O \end{array} \right)$$

has size $n \times m$ (the zero blocks may be absent).



Calculating the Pseudoinverse

Theorem 15.9

Let A be $m \times n$ matrix.

① $A^H A A^+ = A^+ A A^H = A^H,$

② If $r(A) = m \leq n$ then A^+ is a right inverse and can be found as

$$A^+ = A^H (A A^H)^{-1}.$$

③ If $r(A) = n \leq m$ then A^+ is a left inverse and can be found as

$$A^+ = (A^H A)^{-1} A^H.$$



The Least Squares Method

Theorem 15.10

The vector X minimizes $|AX - B|$ if and only if X satisfies the **normal equations**

$$A^H A X = A^H B.$$

The normal equations always have a solution thus such an X always exists. If X_0 is another solution of the normal equations then $X - X_0 \in \text{Ker } A$.

Theorem 15.11

The vector $X_0 = A^+ B$ satisfies the normal equations. Thus all vectors X that minimizes $|AX - B|$ can be described as

$$X = A^+ B + Y, \quad Y \in \text{Ker } A.$$

